Distillation-Constrained Prototype Representation Network for Hyperspectral Image Incremental Classification

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***Abstract*— Oriented to adaptive recognition of the new land-cover categories, incremental classification (IC) that aims to complete adaptive classification with continuous learning is urgent and crucial for hyperspectral image classification (HSIC). Nevertheless, deep-learning-based HSIC models adopted the learning paradigm with fixed classes yield unsatisfactory inference in the situation of IC due to the catastrophic forgetting problem. To eliminate the recognition gap and maintain the old knowledge during IC, in this article, we propose a novel approach called the distillation-constrained prototype representation network (DCPRN) for hyperspectral image IC (HSIIC). The primary goal of DCPRN is to enhance the discriminative capability for recognizing the original classes in HSIIC while effectively integrating both the original and the incremental knowledge to facilitate adaptive learning. Specifically, the proposed framework incorporates a prototype representation mechanism, which serves as a bridge for knowledge transfer and integration between the initial and incremental learning phases of HSIIC. In addition, we present a dual knowledge distillation (DKD) module in incremental learning, which integrates discriminative information at both the feature level (FL) and decision level (DL). In this way, the proposed mechanism enables flexible and dynamic adaptation to new classes and overcomes the limitations of fixed-category feature learning. Extensive experimental analysis conducted on three popular datasets validates the superiority of the proposed DCPRN method compared with other typical HSIIC approaches.**

***Index Terms*— Hyperspectral image incremental classification (HSIIC), knowledge distillation (KD), prototype learning (PL).**

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I. Introduction

HYPERSPECTRAL image (HSI) involves rich spatial–spectral information about the land-cover target that assembles hundreds of continuous bands. HSI classification (HSIC) [1], [2], [3] aims to predict the class label of each pixel and has rendered extensive application in the fields of intelligent agriculture, marine detection, and environmental monitoring [4], [5], [6]. Attribute to the vigorous development of deep learning models, HSIC has made significant progress over the past decade. With the advancement of remote sensing data acquisition capabilities, new categories continue to emerge. As a result, adaptive classification has become increasingly crucial in HSIC due to the growing need for accurate classification for new and previously unseen classes. Nevertheless, HSIC and recognition approaches with deep learning [7], [8], [9], [10] generate unsatisfactory inference results in the recognition of the old categories, i.e., the problem of catastrophic forgetting [11] that is caused by the supervised learning mechanism with the samples from fixed categories. Incremental learning [12], [13], [14], [15] is a beneficial approach for handling new tasks while avoiding catastrophic forgetting of old tasks. Adaptive classification with incremental learning refers to incremental classification (IC), where the classifier is already trained and can recognize new classes without requiring complete retraining of the entire system during the incremental learning phase. Developing the IC paradigm to promote the adaptive recognition ability is effective and beneficial for hyperspectral image IC (HSIIC).

To distinguish the paradigms of different HSIC modes, Fig. 1 presents a simplified flowchart of three representative types of HSIC, including traditional HSIC, cross-scene HSIC, and HSIIC. As can be observed, the traditional HSIC model [16], [17] typically consists of feature extraction and the classifier design. Feature extraction and denoising [18], [19], [20], [21], [22] are responsible for constructing a discriminative representation of the feature space, and the classifier completes the category prediction using predesigned decision criteria. In the situation of cross-scene HSIC [23], [24], the established model intends to implement classification for fixed classes in the target scene with limited labeled samples, and the essential task of cross-scene HSIC is to achieve the consistent expression of spectral–spatial features in the source domain and the target domain. Different from the above classification

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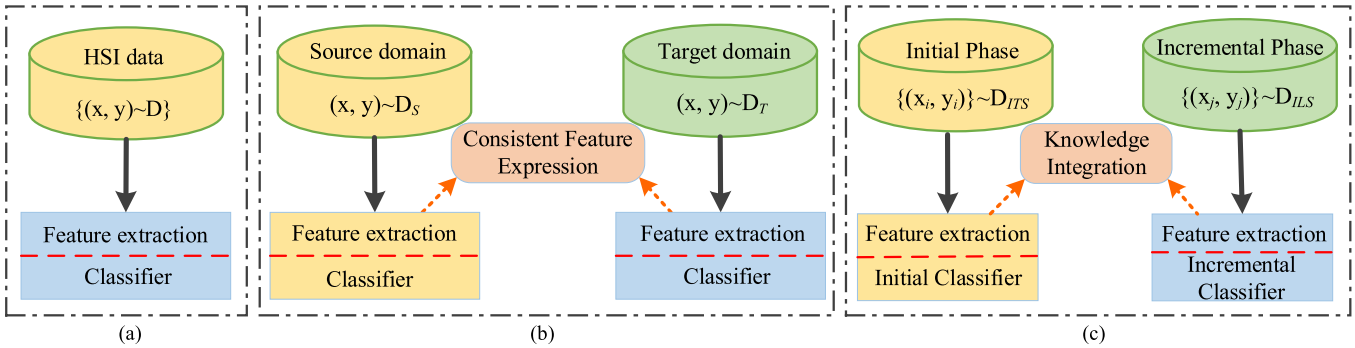


Fig. 1. Illustration of three typical classification paradigms of HSIC. (a) Traditional HSIC. (b) Cross-scene HSIC. (c) HSIIC.

paradigms, HSIIC requires the model to maintain existing cognitive ability during the learning and updating phases with new class samples. The primary objective of HSIIC is to reduce catastrophic forgetting by effectively integrating new knowledge with old knowledge.

Existing incremental learning methods for recognition and classification are categorized into three types: regularization-based solutions, exemplar-based solutions, and dynamic-adjustment-based approaches [25]. First, the regularization-based methods [26], [27] aim to prevent new knowledge from overriding old knowledge by imposing constraints on the new task loss function. For example, the LwF model [28] only uses new data to train the network while retaining the original recognition capability. Second, the exemplar-based approaches [29], [30] select a few old samples to participate in the new knowledge learning with the strategy of sample playback. For example, the iCaRL approach presented a training strategy to alleviate the forgetting problem [31]. However, the sample selection rule for old classes is often not well-designed to handle the variation and instability of IC tasks. Finally, the dynamic-adjustment-based methods [32], [33] aim to update the classification structure by increasing the model capacity to accommodate the increasing new tasks of IC. For instance, the elastic weight consolidation (EWC) method [34] explicitly maintains old classification knowledge by updating the weights of the original tasks.

The recent advancements in incremental learning have led to significant progress in HSIIC [35], [36], [37], [38], [39], [40]. For instance, CILEA-NET [37] utilized a curriculum learning technique to address IC for remote sensing data. The training order of new classes is designed based on the similarity to trained classes. Bai et al. [38] proposed a linear programming incremental learning classifier to learn new capabilities by a trained model with one shot of a new class. Zhao et al. [39] proposed a decoupled structure of the feature representation and classifier to decrease knowledge forgetting and overfitting of HSIIC. Lu et al. [40] proposed a light incremental learning approach that increased a feature transfer module and a classifier to implement incremental learning. Recent studiesabout prototype representation [41], [42] substantially break through the barrier in incremental learning and provide an alternative substitute for storing data or generating samples for the old classes. Notably, the iCaRL approach [31] utilizes prototype representation to derive the exemplar opinion for the solution of training samples in incremental learning. In addition, the prototype augmentation proposed in [43] aims to maintain the decision boundaries of the old task and overcome catastrophic forgetting in incremental learning.

Originally designed for model compression, knowledge distillation (KD) [44], [45] involves building a lightweight model with supervisory information from a larger model. The primary purpose of KD is to utilize a pretrained teacher model with better performance and accuracy to assist in training a student model. In incremental learning applications, KD is employed to transfer knowledge from an initial network (teacher model) to an incremental network (student model) with stable generalization ability. The successful implementation of the LwF method [28], which has applied KD to incremental learning for the first time, has demonstrated the effectiveness of the KD policy in reducing catastrophic forgetting in incremental learning approaches. The technology of KD has been widely applied to the HSIC fields concerned with multiple situations of cross-scene classification [46], [47], few-shot classification [48], [49], and supervised classification [50], [51]. Since the HSIIC has obtained significant attention recently, some experts supplied approaches with KD to address the problem. Among the existing implementations, one typical approach relies on leveraging the exemplars of old classes with KD to enhance the discriminative representation of the initial categories [35], and the other approach transfers knowledge with KD for retaining the previous well-trained information [36] to alleviate the phenomenon of catastrophic forgetting.

While existing incremental learning methods have made some progress in mitigating catastrophic forgetting, there are still challenges that hinder the accuracy of HSIIC. Since the complexity of the HSI data, the primary challenge lies in acquiring a discriminative representation in knowledge integration, which is crucial for the IC model to preserve the semantic representation for the old classes. In addition, the disparities in spectral characteristics and feature distributions between the original and new categories pose challenges to knowledge transfer in IC with KD. To further promote knowledge integration and neglect the effect of the cognitive gap problem caused by the spectral variability of HSI, in this article, a distillation-constrained prototype representation network (DCPRN) for HSIIC is presented to accomplish the effective knowledge integration of the initial and incremental phases. The DCPRN model is composed of three stages: the initial training stage (ITS), the incremental learning stage (ILS), and the prediction stage. The initial training model in

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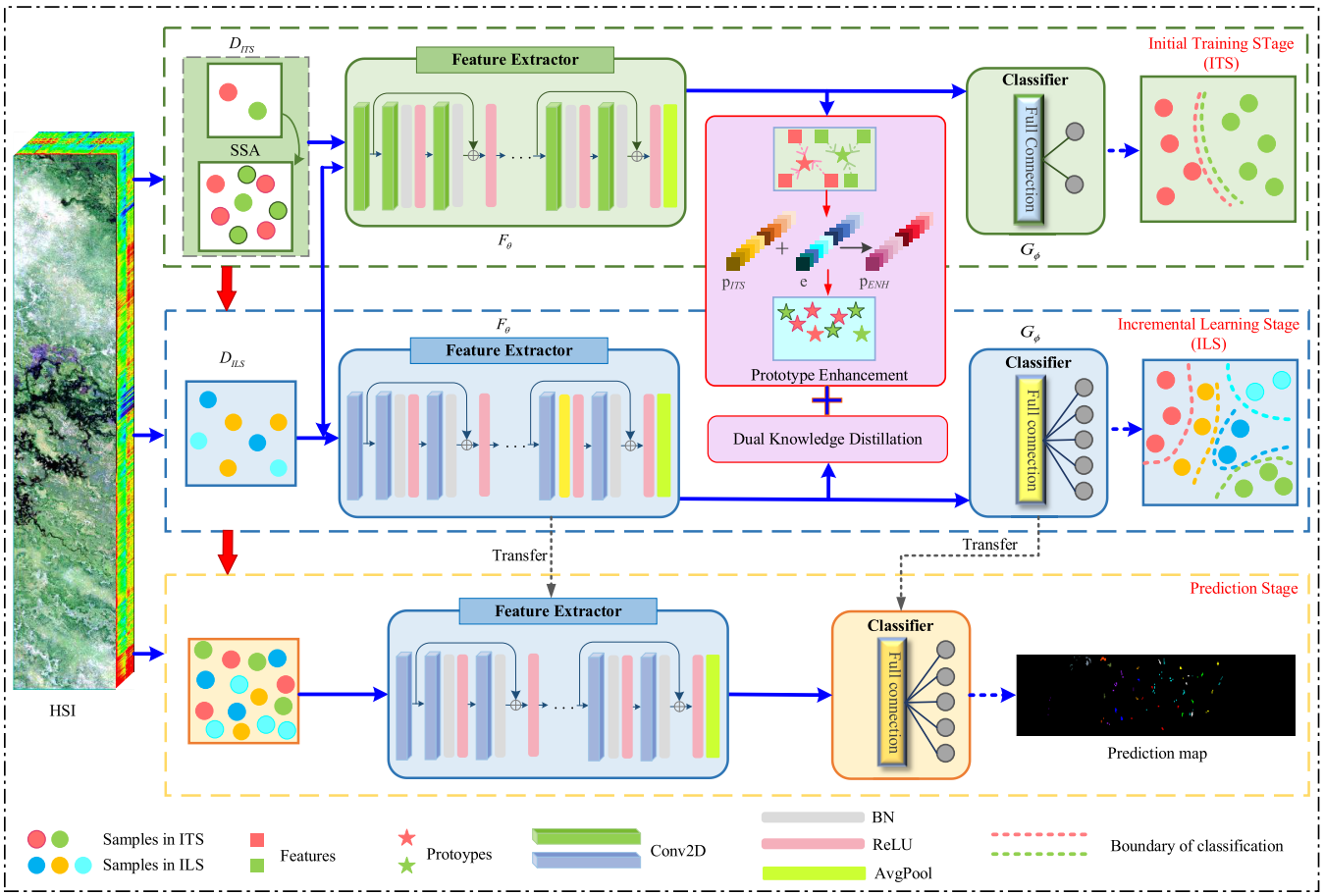


Fig. 2. Flowchart of the proposed DCPRN network. The model consists of sequential stages including the ITS, the ILS, and the prediction stage. First, the initial model is trained with the selected samples of the initial classes for recognition in ITS. Next, ILS aims to build the IC model by completing knowledge integration with the samples from the incremental data. Finally, the test samples are fed into the incremental model for prediction in the prediction stage. Crucially, the PE mechanism is established for enhanced category representation, and the DKD module is constructed to guarantee knowledge transferring and integration.

ITS (ITSM) is built using selected samples from the initial or old training classes for recognition, and an enhanced prototype representation (EPR) mechanism is established for category representation. In the ILS, the incremental learning model (ILSM) learns knowledge by training with samples from the incremental or new training classes and completes knowledge integration simultaneously. In particular, this phase constructs a dual distillation-constrained module to ensure knowledge transfer and integration, eliminating the cognitive gap problem. Finally, the updated classification model is adopted to classify the test samples incrementally in the prediction stage.

The main contributions of this article are summarized as follows.

1) The DCPRN is the first attempt to perform HSIIC, which employs prototype learning (PL) for robust feature representation substituting feature learning with fixed data categories and achieves knowledge integration by the dual distillation module in incremental learning. Notably, the prototype enhancement (PE) mechanism improves the adaptability of the prototype in ILS to alleviate the forgetting of original classification information obtained in ITS.

2) A dual KD (DKD) module is presented to guide knowledge transferring from the initial model to the incremental model. The designed mechanism in ILS contributes to implementing feature-level (FL) and decision-level (DL) knowledge integration simultaneously; in this way, the proposed framework gradually learns a powerful and effective incremental classifier.

3) We designed and conducted comparative experiments with other popular approaches, and a comprehensive analysis demonstrated the proposed framework could effectively improve the generalization ability of HSIIC.

The remainder of this article is organized as follows. In Section II, the details of the proposed DCPRN architecture are described. Section III provides the experimental results and analysis, and conclusions are drawn in Section IV.

II. Proposed Approach

Fig. 2 illustrates the overall architecture of the DCPRN, which includes the three stages mentioned earlier and two primary components: the EPR network and the DKD module. The ITS builds the ITSM with selected samples of initial classes, while the ILS establishes the ILSM by incorporating new training samples of incremental classes and integrating knowledge. In the prediction stage, the trained model is utilized to classify test samples incrementally. The EPR network in DCPRN adopts PL for feature representation in the ITS and ILS periods, which breaks through the restriction of feature modeling with fixed categories. The DKD module effectively

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TABLE I
BACKBONE STRUCTURE OF DCPRN

	layer	Input size	Kernel sets	Output size
Feature Extractor	Conv1	20×9×9	3×3, 64	64×9×9
	Block_1	64×9×9	$\begin{bmatrix}3\times3,64\\3\times3,64\end{bmatrix}$	64×9×9
	Block_2	64×9×9	$\begin{bmatrix}3\times3,128\\3\times3,128\end{bmatrix}$	128×5×5
	Block_3	128×5×5	$\begin{bmatrix}3\times3,256\\3\times3,256\end{bmatrix}$	256×3×3
	Average pool	256×3×3		256×1×1
Classifier	Fully Connection	1×256		1×P in ITS 1×S in ILS

constrains the ILSM to integrate discriminative information and decision knowledge at both the FL and DL. Further details of each component are described in Sections II-B and II-C.

A. Problem Definition

The training phase for HSIIC includes the ITS and ILS, the training set of HSI data $D_{train} = \{D_{ITS}, D_{ILS}\}$ is composed of the initial classes denoted as $D_{ITS} = \cup_{t=1}^{P}\{(x_t^j, y_t^j)\}_{j=1}^{N_t}$ and the incremental classes called as $D_{ILS} = \cup_{t=P+1}^{P+Q}\{(x_t^j, y_t^j)\}_{j=1}^{N_t}$, where N_t denotes the number of samples of the tth class, and (x_t^j, y_t^j) means the jth sample of the tth class. In addition, the number of classes in D_{ITS} and D_{ILS} is P and Q, respectively. For HSIIC, the ITSM is first trained with the samples from D_{ITS} in ITS, and the ILSM is established with the samples from D_{ILS} in ILS. Afterward, all test samples are classified with the updated classifier in the prediction stage.

B. Enhanced Prototype Representation Network

The goal of the prototype representation of the DCPRN is to learn the representative feature space of the given class, which provides concise feature support for the subsequent metric decision of IC. The calculation of each prototype in the DCPRN is composed of the backbone network and PE. Briefly, the backbone network of the ITS and ILS extracts a representative feature space of the specified class. The PE updates and improves the robustness of the adaptability of the initial prototypes in the ILS.

1) Backbone Network: The backbone network consists of a feature extractor F_θ and a unified classifier G_ϕ. We employed the classic and powerful ResNet for the implementation of F_θ. Specifically, F_θ is divided into five main components, and the specific structure is listed in Table I. G_ϕ is implemented with a fully connected layer, with the node number varying dynamically according to the number of classification categories in the ITS and ILS.

Moreover, to improve the discriminative ability of the ITSM, a self-supervised sample augmentation (SSA) is adopted to increase the diversity of the training samples. In specific, the training samples are rotated by 90°, 180°, and 270° to generate an additional three times classification tasks. The augmentation strategy is settled with the following equation:

$$x^* = rotate(x_i, \omega), \omega = \{90°, 180°, 270°\} \tag{1}$$

where x_t is the original training sample and x^* is the sample obtained by rotating x_t by an angle of ω.

Accordingly, the multiclass cross-entropy loss function as shown in the following equation is applied to train ITSM for P-class classification in the ITS:

$$L_{cls} = -\frac{1}{N_P}\sum_{i=1}^{N_P}\sum_{t=1}^{P} y_{it}\log g(x_{it}) \tag{2}$$

where y_{it} denotes the label of the sample x_i and $g(x_{it})$ represents the predicted probability.

2) Prototype Enhancement (PE): After establishing the ITSM, the prototype for each class in D_{ITS} denoted as p_{ITS} is computed by the mean operator of the specified class accordingly. To enhance the adaptive ability of the prototype representation, p_{ITS} is optimized based on the mean and variance of the normal distribution. The specific operation of the prototype update is defined by the following formula, which is adaptive to the input in a training batch during the ILS:

$$p_{ENH} = p_{ITS} + e \tag{3}$$

where p_{ENH} denotes the prototype after PE, $e \sim N(v, \sigma)$ with v represents the mean of the class of p_{ITS}, and σ is the covariance of the specified class accordingly. Subsequently, p_{ENH} are fed into G_ϕ along with the generated features of the samples from D_{ILS} to enhance the incremental learning performance.

C. DKD for ILS

In the proposed DCPRN architecture, we introduce DKD into the HSIIC model to retain previously learned information without catastrophic forgetting by substituting the teacher–student model with the initial incremental model. In particular, the illustration of two KD blocks is displayed in Fig. 3. To alleviate the double mismatch that existed in both F_θ and G_ϕ, dual modules are constructed to accomplish knowledge transferring with the KD both at an FL and DL, which are denoted as KD in FL and KD in DL in this article. Moreover, a KL loss is applied to measure the similarity between the old and new probability distributions. With the combination of transferring from the features and decision response, the DCPRN framework guides ILSM to retain the knowledge of the ITSM.

1) KD in FL: The FL KD constraint encourages the incremental model to capture similar patterns, structures, and discriminative information as the initial model, which enhances the ability to generalize and make accurate predictions on the new classes. Specifically, to implement the KD at the FL, we constrain F_θ by matching the features extracted by the ILSM to the ITSM with the following formula:

$$L_{KD_FL} = -T^2\sum_{t=1}^{Q} p_{ITS}(x_t, T)\log(p_{ILS}(x_t, T)) \tag{4}$$

where $p_{ITS/ILS}(x_t, T) = Soft\max(f_{ITS/ILS}(x_t), T)$ and $f_{ILS}(x_t)$ and $f_{ITS}(x_t)$ denote the features of the training sample x_t of ILSM and ITSM, respectively. Generally, T is set to 2 in Section III.

The KD loss in FL measures the discrepancy between the feature representations of the ILSM and ITSM. By minimizing

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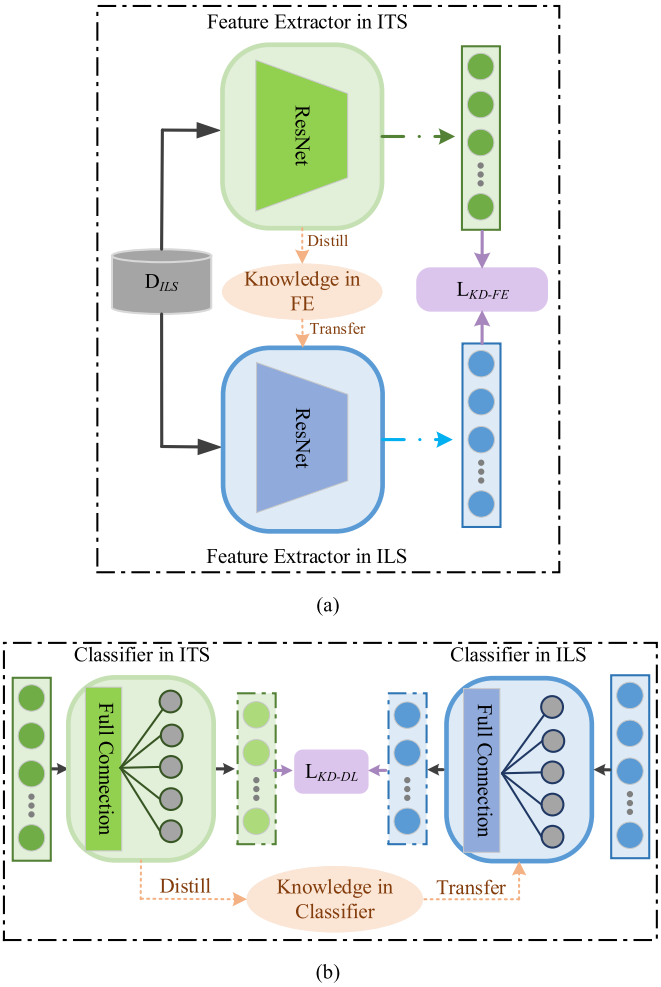


Fig. 3. Diagram of the DKD modules. (a) KD in FL. (b) KD in DL.

this loss, the ILSM learns to align the extracted features with those of the ITSM, effectively transferring knowledge from the initial model to the incremental model.

2) KD in DL: The DL KD constraint encourages the incremental model to make similar predictions as the initial model, which enables the model to benefit from the knowledge and improve performance in the new classes. Specifically, to implement the KD at the DL, we constrain G_ϕ by matching the output of the ILSM with the output of the ITSM with the following equation:

$$L_{KD_DL} = -T^2 \sum_{t,j}^{N_t} q_{ITS}(x_t, T) \log(q_{ILS}(x_t, T)) \quad (5)$$

where $q_{ITS/ILS}(x_t, T) = Soft\max(g_{ITS/ILS}(x_t), T)$ and $g_{ILS}(x_t)$ and $g_{ITS}(x_t)$ denote the output of the samples x_t on G_ϕ of the ILSM and the ITSM, respectively.

The KD loss in DL measures the cross entropy between the soft targets from the initial model and the predicted probabilities from the incremental model. By minimizing this loss, the ILSM learns to mimic the output distribution of the ITSM, effectively transferring the knowledge of class probabilities from the teacher model to the student model.

Eventually, the loss function in the ILS of DCPRN is composed of the cross-entropy loss and the KL dispersion loss, which is defined as follows:

$$L = L'_{cls} + \alpha * L_{KD_FL} + \beta * L_{KD_DL} \quad (6)$$

$$L'_{cls} = -\frac{1}{N} \sum_{i,j}^{N_t} \sum_{t=1}^{P+Q} y_{it} \log g(x_{it}, p_{ENH_{jt}}) \quad (7)$$

where α and β are the coefficients and L'_{cls} is the cross-entropy loss that is calculated in the ILS.

TABLE II
SUMMARY OF THREE HSI DATASETS

Data set	Botswana	Houston	Salinas
Areas	Botswana	Houston, USA	California, USA
Sensor	HYPERION	ITRES CASI-1500	AVIRIS
Spatial Size	256×1476	349×1905	512×614
Num. of Bands	145	144	204
Num. of Classes	14	15	16

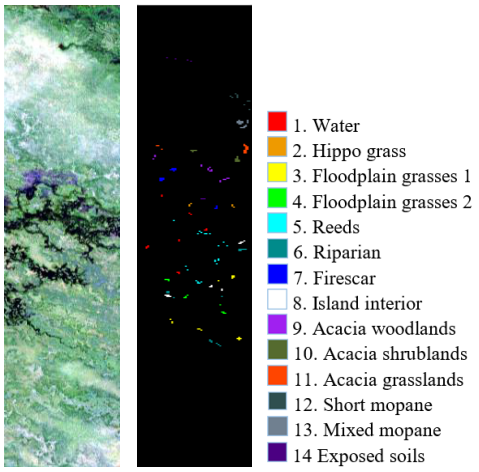


Fig. 4. Data of Botswana. (a) False-color image (bands 10, 67, and 114). (b) Ground-truth image.

III. Experiment and Analysis

A. Data Description

We evaluated the performance of our proposed approach on three widely used HSI datasets. Information on the three datasets is provided in Table II briefly.

1) *Botswana:* The scene captured over the Okavango Delta in 2001–2004, which was acquired by the HYPERION sensor of the EO-1 satellite with 145 spectral bands ranging from 400 to 2500 nm and involves 14 types of ground objects. These data consist of 256 × 1476 pixels, and the false-color and ground-truth images are illustrated in Fig. 4(a) and (b), respectively.

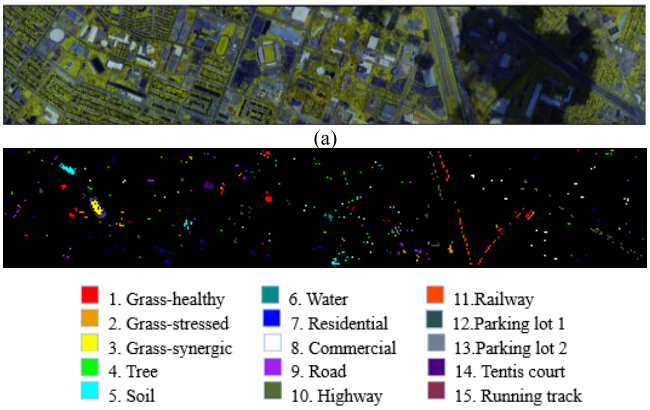
2) *Houston:* The Houston dataset was collected by the ITRES CASI-1500 sensor, which covers the University of Houston campus. The resolution of this image is 349 × 1905 with 144 bands. In specific, the false-color and ground-truth images with 15 classes of land-cover objects are shown in Fig. 5(a) and (b), respectively.

3) *Salinas:* The Salinas dataset was gathered by the 224-band AVIRIS sensor over the Salinas Valley, CA, USA. The scene size of this scene is 512 × 217, and the spatial

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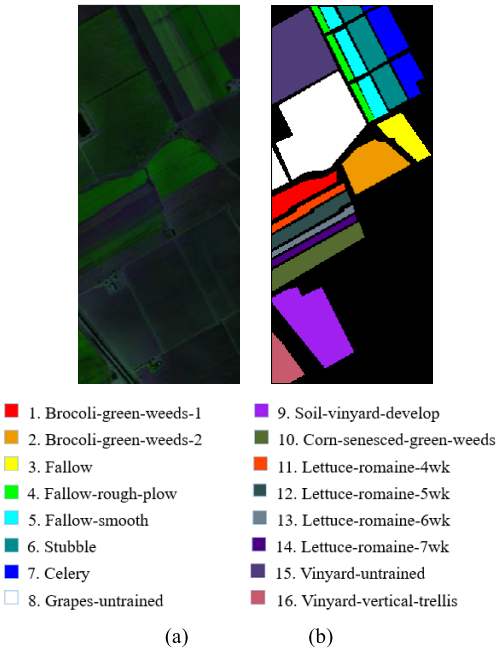
Algorithm 1 Algorithm of DCPRN

Stage	Algorithm of DCPRN
ITS	**Input:** D_{ITS}
	For $n = 1$ to Epoch **do** **For** $b = 1$ to Batch **do** x^* ←SSA in the ITS according to Equ. (1) L_{cls} ← Calculate the loss with Equ. (2) θ, ϕ =←-$\nabla_{\theta,\phi}(L_{cls})$ **End** **End**
	θ_{ITS}, ϕ_{ITS} ← Parameters of F_θ and G_ϕ for ITSM
	For j in$\{1, \ldots, P\}$ do $p_{ITS,j}$ ← Generated with all training features of the jth class. **End**
	Output:θ_{ITS}, ϕ_{ITS}, $\cup_{j=1}^{P}\{p_{ITS,j}\}$
ILS	**Input:** D_{ILS}, $\cup_{j=1}^{P}\{p_{ITS,j}\}$
	$\cup_{j=1}^{P}\{p_{ENH,j}\}$ ← Calculate with Equ. (3)
	For $n = 1$ to Epoch **do** **For** $b = 1$ to Batch **do** L ←Calculate the loss with Equ. (6) θ, ϕ =←-$\nabla_{\theta,\phi}(L)$ **End** **End**
	Output: ILSM with θ_{ILS}, ϕ_{ILS}
Prediction stage	**Input:** Testing samples
	Prediction with ILSM
	Output: Classification results



(a) (b)

Fig. 5. Data of Houston. (a) False-color image (bands 123, 68, and 9). (b) Ground-truth image.



(a) (b)

Fig. 6. Data of Salinas. (a) False-color image (bands 150, 80, and 172). (b) Ground-truth image.

resolution is approximately 3.7 m. After removing the water absorption and low SNR bands, 204 bands remained. The false-color and ground-truth images consist of 16 labeled classes and are displayed in Fig. 6(a) and (b), respectively.

B. Experimental Configuration

All experiments are conducted with PyTorch 1.2 and Python 3.7 equipped with an Nvidia GTX 1050 Ti GPU, an Intel I5-8400 CPU, and 8 GB of RAM.

In the following experiments, the HSI datasets are first sliced to the samples with a size of 9×9 and preprocessed by a principal component analysis (PCA) to obtain the 20 principal components. In addition, the learning rate is set to 0.0001 for the Botswana and Houston datasets and 0.00001 for the Salinas data. Corresponding to the learning rate and convergence speed, we set the number of training epochs to 100 for the Botswana and Houston datasets, and 130 for the Salinas dataset. The batch size is set to 128, and the Adam optimizer is adopted to guide the parameter update optimization. Moreover, for the two coefficient parameters α and β in the loss function, we set both of them to 1 for the Botswana and Houston datasets. For the Salinas dataset, the two parameters are set as $\alpha = 3$ and $\beta = 2$ empirically.

The overall accuracy (OA) and Kappa are employed as objective criteria to evaluate the performance of the approaches. In addition, to further measure the robustness of the HSIIC implementation, we extend four criteria of OA to estimate the effectiveness comprehensively. Specifically, we denoted the OA result of the initial classes with the ILSM as OA-INIL, the accuracy result of the initial classes with the ITSM as OA-INIT, the OA results of the incremental classes with the ILSM as OA-ICIL, and the overall results of the testing data with the ILSM as OA-ILSM. Specifically, OA-INIL assesses the classification performance of how well the initial model performs on the old classes. OA-INII reflects the capability to correctly classify the newly introduced

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TABLE III
COMPARISON OF CLASS PERFORMANCE OF DIFFERENT APPROACHES FOR THE BOTSWANA DATA

	Class	LwF	End2End	EWC	iCaRL	FOSTER	PASS	DCPRN
Initial Classes	Reeds	75.28 ± 2.95	6.13 ± 0.25	33.74 ± 3.91	46.19 ± 9.33	46.99 ± 6.38	**94.87 ± 1.09**	88.62 ± 3.17
	Island interior	41.34 ± 2.19	4.51 ± 0.42	24.06 ± 2.17	31.90 ± 15.53	78.13 ± 8.09	6.21 ± 10.77	**95.57 ± 5.85**
	Acacia woodlands	18.95 ± 11.31	0.16 ± 0.05	22.09 ± 3.41	71.66 ± 9.58	67.77 ± 10.40	83.69 ± 8.82	**93.82 ± 3.52**
	Acacia grasslands	12.26 ± 11.37	2.44 ± 1.17	31.93 ± 3.86	41.64 ± 8.17	83.54 ± 10.27	72.92 ± 37.30	**90.62 ± 6.05**
	Exposed soils	0.81 ± 0.57	1.58 ± 0.34	80.16 ± 6.88	**97.90 ± 1.29**	91.16 ± 2.54	42.32 ± 43.69	74.74 ± 2.86
Incremental Classes	Water	3.47 ± 4.05	34.70 ± 2.31	87.50 ± 4.33	**99.54 ± 0.16**	75.04 ± 3.75	98.81 ± 0.86	99.19 ± 0.54
	Hippo grass	17.10 ± 14.58	8.64 ± 3.69	19.94 ± 4.24	**96.29 ± 1.08**	51.29 ± 3.83	12.08 ± 21.77	61.98 ± 2.05
	Floodplain grasses1	11.80 ± 12.11	25.74 ± 2.49	28.93 ± 1.03	93.23 ± 0.94	66.69 ± 25.40	78.73 ± 16.19	**95.06 ± 5.72**
	Floodplain grasses2	10.84 ± 12.28	40.47 ± 6.98	11.26 ± 1.57	98.60 ± 0.74	92.09 ± 6.96	80.00 ± 16.92	**99.50 ± 0.24**
	Riparian	29.78 ± 16.70	3.20 ± 1.27	31.46 ± 0.29	59.20 ± 12.63	40.60 ± 6.41	54.57 ± 11.03	**72.64 ±2.73**
	Firescar	19.18 ± 15.16	1.62 ± 0.85	28.52 ± 2.41	91.89 ± 2.93	51.51 ± 14.25	99.31 ± 0.86	**99.38 ± 0.31**
	Acacia shrublands	2.90 ± 1.63	2.81 ± 0.09	17.64 ± 2.11	**99.60 ± 0.01**	83.39 ± 8.91	92.50 ± 2.81	99.12 ± 0.65
	Short mopane	7.37 ± 3.43	3.83 ± 0.28	35.98 ± 3.06	**98.07 ± 1.49**	96.24 ± 2.69	82.21 ± 15.19	93.70 ± 3.29
	Mixed mopane	13.16 ± 1.38	1.87 ±0.76	29.76 ± 1.74	81.44 ± 15.79	61.94 ± 9.77	93.88 ± 4.43	**98.88 ± 1.16**
Performance	OA	21.52 ± 2.61	11.27 ± 0.50	43.74 ± 3.84	78.28 ± 3.44	69.54 ± 0.02	76.22 ± 2.86	**92.17 ± 2.01**
	Kappa	14.99 ± 0.03	5.69 ± 0.00	39.18 ± 0.04	76.32 ± 0.04	67.03 ± 0.02	74.19 ± 3.10	**85.06 ± 0.04**

categories using the knowledge originally learned. OA-ICIL measures the performance of the incremental model specifically on the new classes. OA-ILSM evaluates the ability to retain knowledge of the initial categories while learning new classes incrementally. In summary, OA-INIL, OA-INII, OA-ICIL, and OA-ILSM provide specific assessments of different aspects of the classification process in the context of incremental learning. These criteria are utilized to measure the robustness of the approaches and the ability to handle the HSIIC scenarios. Notably, among the four OAs, OA-ILSM is considered the most crucial criterion for evaluation. It is worth mentioning that OA-ILSM is the general OA metric commonly utilized in other papers as well. In the following paper, OA in the tables and figures for results analysis stands for OA-ILSM.

C. Result and Analysis

In this section, we have conducted experiments on the three datasets with different HSIIC approaches. The compared advanced IL methods include LwF [28], iCaRL [31], EWC [34], End2End [52], feature boosting and compression for class incremental learning (FOSTER) [53], and prototype augmentation and self-supervision (PASS) [43]. A brief introduction of the mentioned approaches is listed as follows.

1) *LwF:* The approach utilizes incremental samples and feeds into the incremental and old models. Nodes for the new class are gradually added to the output layer.
2) *End2End:* The work is achieved within an end-to-end framework, where new data and some samples of the old classes are utilized in incremental learning.
3) *EWC:* The method regulates the learning speed of specific weights according to the relevance to the previous tasks, implements incremental tasks, and retains knowledge from the old model.
4) *iCaRL:* The approach involves constructing example sample sets for each class and utilizing class prototype representations throughout the entire incremental process.
5) *FOSTER:* As a dynamic-adjustment-based approach, the model addresses the storage burden of incremental learning by employing distillation to merge the backbone of ITS and ITL at a time.
6) *PASS*: The algorithm is a regularization-based approach, which implements incremental learning by optimizing a single prototype for each incremental class.

Notably, for a more intuitive view and a fair comparison, the categories of the initial training phase are fixed for different methods. The selected ratio of the training set of the three datasets is 10%, and the whole dataset is used as the testing set to verify and present the results. P classes are randomly selected to form the training samples in the ITS and P values are set to 5, 8, and 7 for the Botswana, Houston, and Salinas datasets, respectively. Accordingly, Q is 9, 7, and 9 for the three datasets in ILS. The classification performance is presented in Tables III–V, and the colorful result maps are demonstrated in Figs. 7–9. To ensure a reliable classification result, we perform each method five times with OA computed.

In the experiment for the Botswana data, the initial classes in ITS include Water, Floodplain grasses1, Floodplain grasses2 Firescar, and Acacia shrublands. As illustrated in Table III, the classification accuracy of the DCPRN method is higher than that of other models on nine categories. In particular, the accuracies of the Island interior, Acacia woodlands, and Acacia grasslands are more than 90%, which proves that the presented method alleviates the forgetfulness of old information to a great extent. The accuracies of LwF and End2End yield unsatisfactory results, which demonstrate that the two mentioned approaches are ineffective for HSIIC due to the inherent complexity of the HSI data. Similarly, EWC generates an unideal classification map and accuracy. Compared to iCaRL, the proposed DCPRN improves by 14% and 9% for the OA and Kappa, respectively. Due to the inherent complexity of HSI, FOSTER achieved the performance for the Botswana data, and the OA is 69.54%. PASS demonstrates superior effectiveness than FOSTER. Notably, although the precision of Exposed soils, Hippo grass, and Riparian is below 80%, the performance of the proposed DCPRN model is still the highest of the compared approaches.

For the experiment of Houston data, the initial training sets include Trees, Soil, Water, Residential, Highway, Railway, Parking lot1, and Tennis court. The compared accuracies

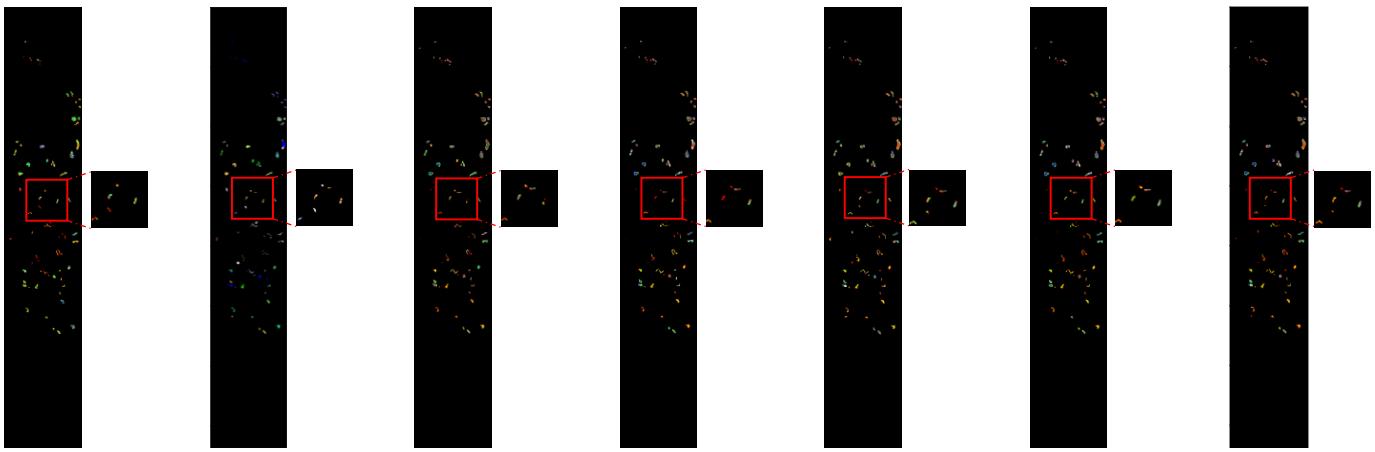
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TABLE IV
COMPARISON OF CLASS PERFORMANCE OF DIFFERENT APPROACHES FOR THE HOUSTON DATA

	Class	LwF	End2End	EWC	iCaRL	FOSTER	PASS	DCPRN
Initial Classes	Tree	37.30 ± 7.40	9.93 ± 12.35	31.75 ± 4.49	33.14 ± 10.41	81.03 ± 1.58	**81.45 ± 2.98**	52.36 ± 6.65
	Soil	72.30 ± 5.29	8.77 ± 8.32	44.23 ± 3.71	57.33 ± 14.05	77.58 ± 4.26	66.02 ± 18.96	**97.30 ± 1.27**
	Water	1.92 ± 0.79	12.02 ± 2.32	4.03 ± 0.16	42.08 ± 8.12	31.63 ± 16.42	5.23 ± 9.14	**78.15 ± 3.76**
	Residential	41.63 ± 3.26	2.20 ± 1.51	7.67 ± 1.08	20.01 ± 11.14	28.38 ± 12.81	27.43 ± 10.87	**80.90 ± 3.37**
	Highway	**84.09 ± 3.35**	9.01 ± 2.56	18.97 ± 2.68	26.75 ± 1.65	26.59 ± 0.803	33.84 ± 26.78	74.70 ± 6.34
	Railway	41.98 ± 4.17	3.32 ± 0.75	15.65 ± 2.21	24.64 ± 10.23	33.07 ± 25.66	3.79 ± 4.38	**76.08 ± 2.43**
	Parking lot1	64.66 ± 3.12	15.76 ± 2.24	10.85 ± 1.53	47.73 ± 12.31	30.38 ± 10.44	61.52 ± 6.34	**65.55 ± 8.26**
	Tentis court	22.02 ± 8.13	15.22 ± 1.93	10.73 ± 1.75	84.81 ± 2.82	83.88 ± 5.79	28.46 ± 29.81	**94.06 ± 1.55**
Incremental Classes	Grass-healthy	9.37 ± 7.91	33.55 ± 3.93	66.22 ± 4.68	**94.40 ± 6.24**	79.07 ± 3.12	82.94 ± 11.44	82.61 ± 4.75
	Grass-stressed	15.37 ± 5.33	1.50 ± 1.93	28.97 ± 3.96	85.55 ± 4.12	42.94 ± 24.89	**88.28 ± 7.44**	84.60 ± 3.72
	Grass-synergic	1.29 ± 0.96	3.49 ± 6.58	40.11 ± 3.21	**98.32 ± 0.78**	92.86 ± 1.76	97.42 ± 2.37	93.46 ± 1.18
	Commercial	11.50 ± 9.14	18.94 ± 2.59	49.03 ± 3.98	61.38 ± 6.62	55.77 ± 12.45	**82.17 ± 2.32**	67.54 ± 3.90
	Road	2.92 ± 3.11	4.68 ± 5.12	15.38 ± 2.17	**93.73 ± 1.91**	32.24 ± 14.81	85.91 ± 3.04	85.46 ± 2.90
	Parking lost2	11.99 ± 1.32	3.63 ± 2.86	23.72 ± 3.35	84.81 ± 2.83	54.20 ± 14.87	**86.99 ± 5.77**	84.61 ± 4.52
	Running track	1.70 ± 0.95	20.56 ± 2.37	15.84 ± 2.24	86.25 ± 4.98	97.56 ± 0.74	**98.61 ± 1.63**	85.58 ± 7.39
Performance	OA	34.18 ± 0.93	13.80 ± 0.86	28.08 ± 1.51	56.69 ± 2.77	54.06 ± 2.94	63.35 ± 3.86	**78.46 ± 1.91**
	Kappa	29.02 ± 0.01	6.37 ± 0.01	23.11 ± 0.18	53.62 ± 0.03	50.52 ± 3.23	60.31 ± 4.21	**77.58 ± 0.02**

TABLE V
COMPARISON OF CLASS PERFORMANCE OF DIFFERENT APPROACHES FOR THE SALINAS DATA

	Class	LwF	End2End	EWC	iCaRL	FOSTER	PASS	DCPRN
Initial Classes	Brocoli-green-weeds-1	63.66 ± 3.80	7.10 ± 4.04	75.00 ± 4.30	68.59 ± 1.78	67.65 ± 39.32	1.44 ± 2.89	**95.01 ± 6.15**
	Brocoli-green-weeds-2	0.54 ± 0.22	6.19 ± 6.35	34.10 ± 3.75	56.49 ± 1.25	41.25 ± 33.12	35.73 ± 43.77	**77.17 ± 2.67**
	Fallow-rough-plow	59.54 ± 2.57	9.97 ± 0.99	54.73 ± 1.49	86.76 ± 1.90	**95.01 ± 2.78**	57.66 ± 37.54	93.10 ± 3.57
	Soil-vinyard-develop	44.71 ± 3.59	5.01 ± 0.59	51.61 ± 4.06	70.34 ±3.24	80.07 ±31.39	0.16 ±0.32	**83.86 ± 1.92**
	Corn-senesced-green-weeds	40.55 ± 3.79	3.87 ± 1.30	18.97 ± 3.25	26.19 ± 1.63	76.58 ± 5.38	76.43 ± 6.03	**86.40 ±1.12**
	Lettuce-romaine-5wk	6.74 ±3.28	17.07 ± 2.41	33.15 ± 3.75	71.77 ±1.22	89.03 ±15.41	5.93 ±9.66	**91.69 ± 4.02**
	Vinyard-vertical-trellis	16.14 ± 0.78	4.98 ± 0.69	22.14 ± 2.84	43.41 ± 1.74	56.70 ± 23.77	24.22 ± 30.01	**58.58 ± 1.96**
Incremental Classes	Fallow	0.17 ± 0.24	73.38 ± 3.71	24.00 ± 4.15	74.66 ± 2.02	1.57 ± 2.32	**92.52 ± 5.85**	83.90 ± 1.24
	Fallow-smoth	12.47 ± 1.46	62.90 ± 3.09	48.85 ± 4.88	89.73 ± 4.01	40.54 ± 21.30	**94.17 ± 5.64**	93.10 ± 4.22
	Stubble	90.25 ± 4.40	9.83 ± 2.45	25.00 ± 2.34	99.68 ± 0.26	84.04 ± 4.32	99.26 ± 0.58	**99.99 ± 0.01**
	Celery	85.59 ± 8.23	10.54 ± 6.42	24.94 ± 4.31	97.31 ± 0.72	82.84 ± 4.42	98.82 ± 2.11	**99.77 ± 0.40**
	Grapes-untrained	87.34 ± 3.56	52.01 ± 3.69	47.42 ± 4.71	92.78 ± 2.15	72.05 ± 20.71	85.22 ± 8.02	**95.51 ± 1.72**
	Lettuce-romaine-4wk	0.53 ± 0.24	52.35 ± 3.94	39.19 ± 1.24	67.07 ± 3.10	0.13 ± 0.17	**93.05 ± 2.93**	42.66 ± 3.35
	Lettuce-romaine-6wk	1.86 ± 2.55	13.44 ± 1.23	24.92 ± 4.31	94.39 ± 3.36	81.42 ± 14.28	**98.32 ± 1.76**	58.16 ± 2.56
	Lettuce-romaine-7wk	1.12 ± 0.03	10.30 ± 1.74	24.91 ± 1.31	89.94 ± 2.01	2.07 ± 3.02	**99.76 ± 0.40**	83.40 ± 1.37
	Vinyard-untrained	10.90 ± 4.43	22.18 ± 1.62	25.76 ± 3.18	**95.71 ± 1.10**	22.15 ± 23.06	60.61 ± 30.71	80.92 ±1.19
Performance	OA	45.31 ± 2.28	15.21 ± 2.62	41.93 ± 3.00	83.70 ± 1.02	58.83 ± 7.49	62.83 ± 6.27	**85.40± 2.03**
	Kappa	37.26 ± 0.03	18.98 ± 0.13	33.09 ± 0.04	82.41 ± 0.01	53.86 ± 7.49	58.93 ± 7.07	**84.47 ± 0.02**



(a) (b) (c) (d) (e) (f) (g)

Fig. 7. Classification maps obtained via different HSIIC methods for the Botswana data. (a) LwF. (b) End2End. (c) EWC. (d) iCaRL. (e) FOSTER. (f) PASS. (g) DCPRN.

are listed in Table IV, and the final classification maps are illustrated in Fig. 8. As can be observed, the LwF, End2End, and EWC fail on the recognition of the initial classes, whereas the OA and Kappa of the DCPRN method are the highest in the compared approaches, especially for the initial classes. Precisely, the proposed method generates a better classification

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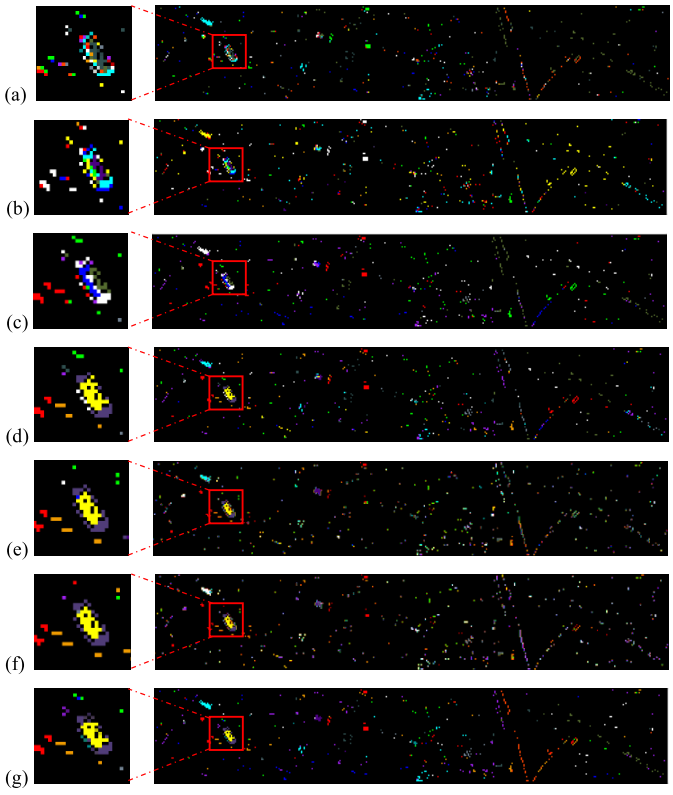


Fig. 8. Classification maps obtained via different HSIIC methods for the Houston data. (a) LwF. (b) End2End. (c) EWC. (d) iCaRL. (e) FOSTER. (f) PASS. (g) DCPRN.

performance than the iCaRL and PASS for most categories. It is worth highlighting that the DCPRN method demonstrates superior performance compared to other approaches in the case of Botswana and Houston datasets. The main reason is that the two datasets lack prominent spatial features, which makes the compared methods that are designed for the natural image with adequate spatial information insufficient to capture the unique characteristics. In contrast, our approach utilizes prototypes for HSI spatial–spectral feature assembling and improves the effectiveness of knowledge transfer during incremental learning. Besides, our method employs constrained distillation at both the FL and DL to ensure the consistency of complex knowledge transfer in HSIIC and mitigate catastrophic forgetting.

The last experiment is performed on the Salinas dataset, and in the ITS phase, seven classes, including Brocoli-green-weeds-1, Brocoli-green-weeds-2, Fallow-rough-plow, Fallow-rough plow, Soli-vineyard-develop, Corn-senesced-green-weeds, Lettuce-romaine-5wk, and Vinyard-vertical-trellis, are selected as the initial classes. As listed in Table V, the End2End approach has the worst performance since the recognition forgetting of the initial classes. The LwF and EWC yield OAs of 45.31% and 41.93%, respectively. The OAs generated with the FOSTER and PASS methods are 58.83% and 62.83%, respectively. The DCPRN model outperforms other methods on all the initial classes, and the values of OA on all five initial classes are higher than 80% except for Brocoli-green-weeds-2 and Vinyard-vertical-trellis, which indicates that the presented method maintains the old classification knowledge obtained in ITS and mitigate the catastrophic forgetting of ILS. Although the iCaRL generates the best OA of Lettuce-romaine-4wk, Lettuce-romaine-6wk, and Lettuce-romaine-7wk, it is still inferior in recognizing the initial classes, especially for the Corn-senesced-green-weeds, which demonstrates that the proposed DCPRN is generally superior to other HSIIC approaches.

D. Performance Analysis With Random Selection in ITS

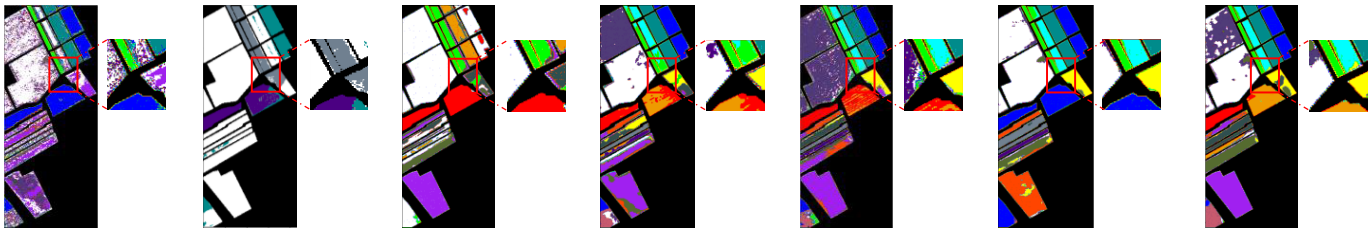
To evaluate the performance of the DCPRN model objectively, in this section, we analyze the compared OAs with random selection on the three datasets. Different from the previous subsection, the categories are randomly selected rather than fixed categories in the ITS and ILS phases, we record the following accuracies for comparison with all the related methods. All the accuracies are comprehensively shown in Fig. 10. As can be seen from all the line graphs in Fig. 10(a), DCPRN outperforms the other models on the four results of OA-INIL, OA-INIT, OA-ICIL, and OA-ILSM on the Botswana dataset. In particular, DCPRN is much higher than the other methods on OA-INIL, and the results represent the degree of old knowledge retention in the incremental process. The precision of the presented method on OA-INIT and OA-ICIL indicates that the present model is well-trained in both stages. Thus, the fold line in Fig. 10 fully demonstrates that the proposed DCPRN method learns knowledge of the new categories by retaining the old knowledge simultaneously, alleviating the catastrophic forgetting problem of IL. The results for the Houston dataset are illustrated in Fig. 10(b). The performance of the DCPRN method is slightly lower than the iCaRL method on OA-ICIL, while the value of the iCaRL method is less than 40% on OA-INIL. The main reason is that the iCaRL approach focuses on the new classification task and ignores the memory retention of the old knowledge. The results of the DCPRN method on OA-INIT, OA-ILSM, and OA-INIL outperformed the other approaches. In particular, the advantage of OA-ILSM accurately illustrated the balance and superiority of the DCPRN method in integrating old and new knowledge, and OA-INIL is higher than the other four implementations by 47.36%, 69.15%, 62.80%, and 41.71%. Similarly, as shown in Fig. 10(c), the DCPRN model has significant advantages over other comparisons for OA-INIT, OA-ILSM, and OA-INIL on the Salinas dataset. Only the accuracy on OA-ICIL is slightly lower than the result of the iCaRL method.

In conclusion, the experimental analysis demonstrates that our proposed method greatly alleviates the forgetting problem of old knowledge in the ITS and effectively integrates old and new information between ITS and ILS for HSIIC.

E. Ablation Study

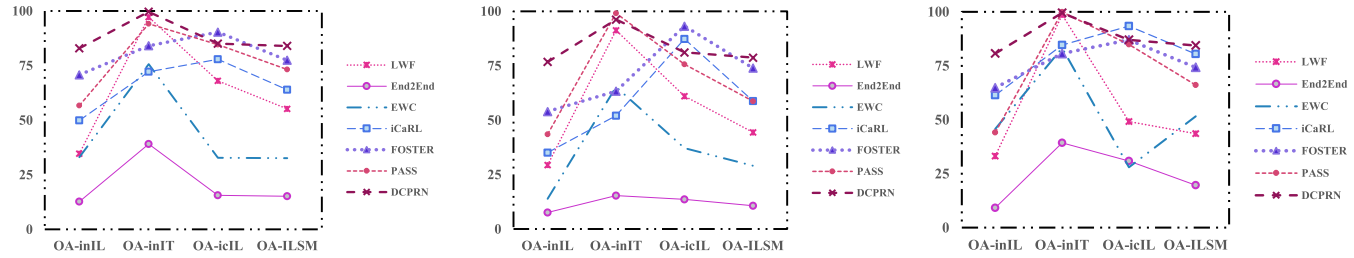
We performed ablation studies on the three datasets to demonstrate the contribution of the different modules in the DCPRN model. In this experiment, the DCPRN without SSA, PE, and DKD is utilized as a baseline for IC implementation, and Table VI presents the specific accuracies of the approaches with different modules. As demonstrated in the table, each module proposed in DCPRN affects the OA-ILSM and OA-INIL. Specifically, compared to the baseline, the addition of the SSA in the ITS resulted in

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(a) (b) (c) (d) (e) (f) (g)

Fig. 9. Classification maps obtained via different HSIIC methods for the Salinas data. (a) LwF. (b) End2End. (c) EWC. (d) iCaRL. (e) FOSTER. (f) PASS. (g) DCPRN.



(a) (b) (c)

Fig. 10. Compared OAs generated with the random selection of the initial class for the three datasets. (a) Botswana. (b) Houston. (c) Salinas.

TABLE VI
ABLATION STUDY ON THE THREE HSI DATASETS

	Module				Accuracy	
	Baseline	SSA	EPR	DKD	OA-ICIL	OA-ILSM
Botswana	√				99.14	64.79
	√	√			98.00	62.71
	√		√		99.66	62.65
	√			√	98.60	70.41
	√	√	√		96.37	64.21
	√	√		√	99.02	72.00
	√		√	√	73.79	77.76
	√	√	√	√	85.09	84.03
Houston	√				97.70	46.30
	√	√			94.13	47.81
	√		√		97.34	47.68
	√			√	96.36	58.69
	√	√	√		92.90	49.83
	√	√		√	96.12	53.76
	√		√	√	89.79	75.87
	√	√	√	√	81.16	78.75
Salinas	√				99.89	57.62
	√	√			99.95	63.08
	√		√		99.10	69.66
	√			√	97.58	73.06
	√	√	√		97.66	68.76
	√	√		√	98.21	65.58
	√		√	√	77.15	69.43
	√	√	√	√	87.05	84.43

varying degrees of improvement in OA-INIT on all three datasets, indicating that the SSA module is beneficial to the model. Moreover, the results demonstrate that the approach of any two modules outperforms the individual modules on OA-INIL for the Botswana and Salinas datasets, and for the Houston data, the accuracies of SSA + DKD on OA-INIL and OA-ILSM are slightly lower than those of the single DKD module. In addition, the results show that the implementation of EPR + DKD improves OA-INIL for the Salinas dataset. The experimental results illustrate that the proposed modules greatly reduce catastrophic forgetting and achieve the integration of old and new knowledge without affecting the IL of new classes. Furthermore, as shown in the last row for each HSI data, it can also be noticed that the approach with the three modules is remarkable for the performance.

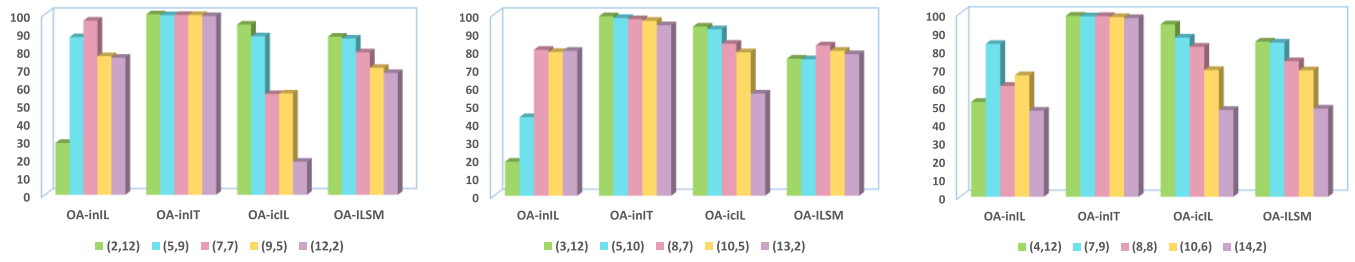
F. Impact of the Class Number in the ITS

In this section, we investigate the effect of setting the different number of categories for the ITS and the ILS. To illustrate the results more visually, we utilize the form of (P, Q) to denote the number of categories in two stages. In specific, we set (2, 12), (5, 9), (7, 7), (9, 5), and (12, 2) for Botswana, (3, 12), (5, 10), (8, 7), (10, 5), and (13, 2) for Houston, and (4, 12), (7, 9), (8, 8), (10, 6), and (14, 2) for Salinas. The specific results of OA-INIL, OA-INIT, OA-ICIL, and OA-ILSM are shown in Fig. 11. As shown in the plots, the number of different P has a minor effect on the OA-INIT for all three datasets, which indicates the satisfactory robustness of the proposed model. OA-ILSM in Botswana demonstrates that the model generates a negative association with the value of P and is positively proportional to the value of Q. For Houston, the impact is smaller than the Botswana data. Specifically, the forgetting problem of old knowledge for Botswana appears more severe with the situation of less Q, as shown in Fig. 11(a). The presence of old and new knowledge is more equilibrated in the setting of (5, 9) than in other conditions. As shown in Fig. 11(b), more Q generates the smaller OA-INIL, which indicates that the number of P is inversely proportional to the knowledge forgetting in ILS. Besides, the classification results of the Salinas with different category numbers are illustrated in Fig. 11(c), and the DCPRN model generates the highest OA-INIL with P of 7 and Q of 9.

G. Analysis of Experimental Parameters

Considering that the sample size is particularly crucial for HSIIC, we conducted experiments with different patch sizes

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(a) (b) (c)

Fig. 11. Accuracy results generated with the DCPRN with P and Q in the DCPRN for the three datasets. (a) Botswana. (b) Houston. (c) Salinas.

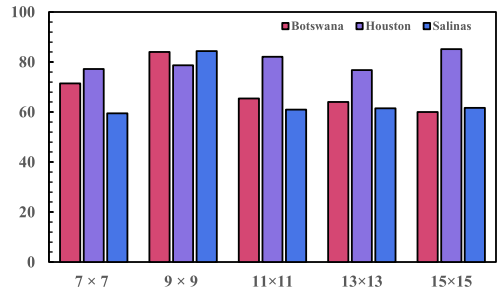


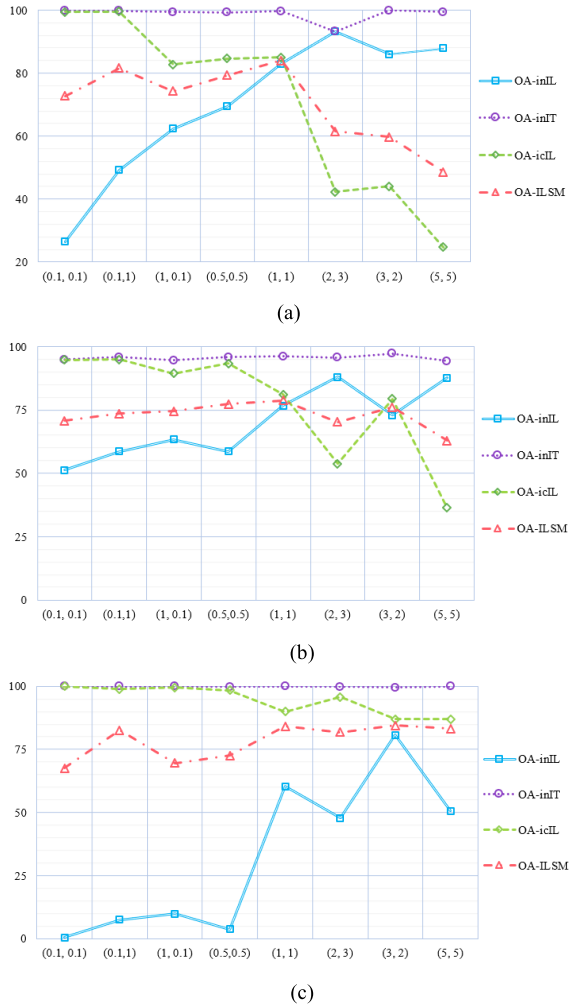
Fig. 12. Performance comparison for the three datasets with different sample sizes.

TABLE VII
COMPARISON OF TIME CONSUMPTION OF DIFFERENT APPROACHES (S)

DATA	STAGE	LwF	End2End	EWC	iCaRL	FOSTER	PASS	DCPRN
Botswana	Train	165.68	712.40	69.03	10.69	40.77	48.52	66.94
	Test	19.43	2.89	1.43	2.88	8.13	2.52	3.16
Houston	Train	307.96	1196.43	77.23	62.89	160.18	194.40	330.24
	Test	40.43	13.66	2.02	12.05	32.01	7.99	6.95
Salinas	Train	627.46	1974.77	93.67	1558.83	2352.559	782.21	520.04
	Test	70.88	42.93	2.02	39.48	69.11	26.26	17.46

ranging from 7 × 7 to 15 × 15. In addition, we provided an analysis of the experimental results. The comparison results are illustrated in Fig. 12. As can be observed, the highest accuracy for the Botswana and Salinas datasets is achieved with a sample size of 9 × 9. Conversely, for the Houston dataset, the highest accuracy is obtained with a sample size of 15 × 15.

Besides, we analyzed the effects of the coefficients of α and β in the DCPRN in this section. The coefficients of (α, β) are set to (0.1, 0.1), (0.1, 1), (1, 0.1), (0.5, 0.5), (1, 1), (2, 3), (3, 2), and (5, 5), and the experimental results are illustrated in Fig. 12. As can be observed, the different settings of (α, β) have little effect on the OA-INIT of the three datasets, and the main reason is that OA-INIT denotes the accuracy of the ITSM, while α and β participated in the ILSM. In addition, OA-ICILs for all three datasets gradually decrease, and the OA-INILs accelerate with the increase of α and β. Specifically, for the Botswana and Houston datasets, OA-ILSMs enhance first and fall later as illustrated in Fig. 13(a) and (b), and the OA-ILSMs reach the highest value when α and β are equal to 1. The OA-ICIL and OA-INIL maintain a relatively higher accuracy with the setting of (1, 1) than in other settings. Due to the mentioned empirical investigation, both α and β are fixed to 1 to yield satisfactory results for Botswana and Houston in the other experiments of this article. Based on a similar observation as shown in Fig. 13(c), the OA-ILSM of the Salinas dataset rises first and reaches a dynamic steady



(a) (b) (c)

Fig. 13. Accuracy results with different α and β and values in the DCPRN for the three datasets. (a) Botswana. (b) Houston. (c) Salinas.

state when α and β increase. In particular, OA-ILSM and OA-INIL are highest along with OA-ICIL close to 90% with the settings of (2, 3). Therefore, for the Salinas dataset, the coefficients of α and β are set to 2 and 3, respectively, in the other experiments to generate pleasing results. As shown in the figure, the OA-ILSM refers to the overall performance during the testing phase by conducting experiments on both the old and new classes. To clarify, although we listed four OAs corresponding to the four lines in Fig. 13, it is important to note that the red line accurately represents the performance of the proposed HSIIC model, which is stable for the three datasets on the different settings of the two parameters.

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TABLE VIII
CLASSIFICATION ACCURACY OF DIFFERENT GROUND OBJECTS IN THE CROSS-SCENE EXPERIMENT

Situation 1			Situation 2			Situation 3		
	Class	Accuracy		Class	Accuracy		Class	Accuracy
Initial Classes (14 classes from Botswana)	Water	80.15 ± 38.96	Initial Classes (15 classes from Houston)	Grass-healthy	36.92 ± 26.72	Initial Classes(7 classes from Houston)	Grass-healthy	57.67 ± 28.83
	Hippo	10.10 ±11.60		Grass-stressed	84.24 ± 3.51		Grass-stressed	21.96 ± 9.48
	Floodplain grasses1	95.54 ± 4.03		Grass-synergic	69.13 ± 26.76		Grass-synergic	98.85 ± 0.87
	Floodplain grasses2	97.49 ± 2.71		Tree	59.79 ± 31.23		Tree	77.73 ± 16.45
	Reeds	71.77 ± 11.49		Soil	75.48 ± 15.62		Soil	95.80 ± 4.05
	Riparian	81.41 ± 8.07		Water2	86.34 ± 9.50		Water2	81.17 ± 11.73
	Firescar	19.77 ± 38.96		Residential	86.51 ± 12.14		Residential	68.77 ± 13.99
	Island interior	100 ± 0		Commercial	53.75 ± 15.46	Incremental Classes(other 8 classes from Houston and all classes from Botswana)	Commercial	71.95 ± 10.85
	Acacia woodlands	87.74 ± 9.09		Road	68.48 ± 28.81		Road	89.11 ± 1.97
	Acacia shrublands	74.27 ± 4.04		Highway	19.38 ± 17.00		Highway	92.83 ± 3.70
	Acacia grasslands	87.87 ± 6.65		Railway	22.19 ± 22.21		Railway	90.01 ± 5.26
	Short mopane	20.99 ± 36.30		Parking Lot1	12.17 ± 5.17		Parking Lot1	89.17 ± 11.12
	Mixed mopane	89.93 ± 15.72		Parking Lot2	69.00 ± 23.66		Parking Lot2	82.30 ± 2.03
	Exposed soils	90.11 ± 1.95		Tennis Court	98.41 ± 1.82		Tennis Court	95.56 ± 1.65
Incremental Classes (15 classes from Houston)	Grass-healthy	90.63 ± 6.74		Running Track	99.36 ± 1.00		Running Track	99.97 ± 0.06
	Grass-stressed	87.67 ± 2.32	Incremental Classes (14 classes from Botswana)	Water	28.67 ± 32.92		Water	93.48 ± 3.33
	Grass-synergic	85.91 ± 18.08		Hippo	50.89 ± 29.12		Hippo	39.80 ± 32.31
	Tree	90.42 ± 0.31		Floodplain grasses1	75.06 ± 13.89		Floodplain grasses1	87.41 ± 8.81
	Soil	99.16 ± 0.36		Floodplain grasses2	93.12 ± 4.77		Floodplain grasses2	95.81 ± 7.68
	Water2	21.79 ± 25.05		Reeds	74.50 ± 10.00		Reeds	69.81 ± 11.33
	Residential	76.55 ± 13.24		Riparian	68.10 ± 7.49		Riparian	47.73 ± 19.20
	Commercial	58.26 ± 11.67		Firescar	39.31 ± 32.21		Firescar	97.92 ± 1.63
	Road	83.34 ± 2.35		Island interior	73.99 ± 34.99		Island interior	92.71 ± 9.20
	Highway	77.36 ± 4.91		Acacia woodlands	91.85 ± 6.56		Acacia woodlands	93.89 ± 4.72
	Railway	71.64 ± 5.55		Acacia shrublands	64.52 ± 24.97		Acacia shrublands	80.65 ± 3.10
	Parking Lot1	87.02 ± 7.00		Acacia grasslands	84.59 ± 3.82		Acacia grasslands	91.80 ± 7.43
	Parking Lot2	81.19 ± 2.28		Short mopane	34.70 ± 33.95		Short mopane	70.94 ± 3.02
	Tennis Court	86.64 ± 15.20		Mixed mopane	71.05 ± 35.73		Mixed mopane	87.09 ± 9.00
	Running Track	98.73 ± 2.03		Exposed soils	60.63 ± 26.69		Exposed soils	73.05 ± 10.89
Performance	OA	80.78 ± 2.23	Performance	OA	59.19 ± 7.37	Performance	OA	79.20 ± 3.10
	Kappa	79.73 ± 2.35		Kappa	57.30 ± 7.63		Kappa	78.06 ± 3.26

Furthermore, we recorded and analyzed the time costs of all the methods. Notably, the results were computed as an average of five iterations. As observed in Table VII, despite not having the lowest time cost, the DCPRN method exhibits remarkable consistency for all three datasets. In contrast, iCaRL generates the highest efficiency for Botswana and Houston, while it exhibits a longer runtime for the Salinas dataset. PASS requires less training time compared to our approach on the Botswana and Houston datasets, yet it incurs more testing time costs. Conversely, our method demonstrates higher efficiency in both training and testing time on the Salinas dataset. Significantly, the proposed method achieves the best accuracy compared to other methods, which indicates that the DCPRN method is competitive in terms of time efficiency and performance.

H. Analysis of Cross-Scene HSIIC

To analyze the performance of the DCPRN model in cross-scene applications, we conducted experiments that involved incorporating initial classes and incremental classes from different datasets. In this section, we utilized the Houston and Botswana datasets and established three scenarios to evaluate the performance of the proposed method under different conditions. Specifically, in Situation 1, we selected initial classes from the Botswana dataset and incremental classes from the Houston dataset. In Situation 2, we reversed the selection with initial classes from the Houston dataset and the incremental classes from the Botswana dataset. In Situation 3, a subset of seven classes from the Houston dataset served as the initial classes, while the incremental classes included the remaining eight classes from Houston and all the classes from the Botswana dataset. The experimental settings for all the experiments include an epoch of 200, a sample size of 9 × 9, a batch size of 128, a learning rate of 0.0001, and a fixed tradeoff parameter of 1. Each experiment in a specific situation was repeated five times, and the classification performance is reported in Table VIII. As observed, our method consistently delivers stable performance across all situations. The accuracy achieved in Situation 1 was 80.78%, and our method yielded an OA of 79.20% in Situation 3. Although there are some classes with unsatisfactory accuracies, such as Hippo, Firescar, and Short mopane in Situation 1, Highway, Railway, and Parking Lot1 in Situation 2, and Grass-stressed in Situation 3, the DCPRN method demonstrates effectiveness and reliability both for most initial classes and incremental classes. In summary, the relevant results show the robustness of our method and the ability to adapt to varying scenarios and datasets in cross-scene applications.

IV. CONCLUSION

In this article, a prototype representation architecture constrained by KD is proposed to accomplish effective knowledge integration for HSIIC. The proposed model mainly consists of a class PE module and a DKD block, which jointly contribute to eliminating the recognition gap and maintaining

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the previously trained knowledge. In particular, the prototype representation network with enhanced augmentation achieves memory retention in the incremental learning phase. The DKD module completes knowledge integration at the FL and DL explicitly. Extensive experimental results and analysis demonstrate that our proposed framework achieves reliable effect and robustness for HSIIC on three public datasets. Our method encounters challenges when dealing with highly imbalanced class distributions, particularly with few samples of new classes from incremental datasets. In the future, we aim to investigate the mentioned issue by designing a constrained strategy that carefully balances the preservation of prior knowledge with the accommodation of new information. In addition, by incorporating attention mechanisms and hybrid approaches into the incremental learning model, the HSIIC model further enhances the overall classification performance in complex spectral variation situations.

REFERENCES

[1] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[2] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang, and M. Song, "MSTNet: A multilevel spectral–spatial transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532513.

[3] H. Yang, H. Yu, K. Zheng, J. Hu, T. Tao, and Q. Zhang, "Hyperspectral image classification based on interactive transformer and CNN with multilevel feature fusion network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[4] W. Sun et al., "A simple and effective spectral–spatial method for mapping large-scale coastal wetlands using China ZY1–02D satellite hyperspectral images," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 104, Dec. 2021, Art. no. 102572.

[5] K. Liu et al., "Mapping coastal wetlands using transformer in transformer deep network on China ZY1–02D hyperspectral satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3891–3903, 2022.

[6] Y. Yuan et al., "Multi-resolution collaborative fusion of SAR, multispectral and hyperspectral images for coastal wetlands mapping," *Remote Sens.*, vol. 14, no. 14, p. 3492, Jul. 2022.

[7] C. Yu, J. Huang, M. Song, Y. Wang, and C.-I. Chang, "Edge-inferring graph neural network with dynamic task-guided self-diagnosis for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535613.

[8] L. Zhang, L. Song, B. Du, and Y. Zhang, "Nonlocal low-rank tensor completion for visual data," *IEEE Trans. Cybern.*, vol. 51, no. 2, pp. 673–685, Feb. 2021.

[9] Y. Wang, X. Chen, E. Zhao, and M. Song, "Self-supervised spectral-level contrastive learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510515.

[10] Y. Wang, X. Chen, F. Wang, M. Song, and C. Yu, "Meta-learning based hyperspectral target detection using Siamese network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527913.

[11] R. K. Pathak and V. Yadav, "Improvised progressive neural network (iPNN) for handling catastrophic forgetting," in *Proc. Int. Conf. Electron. Sustain. Commun. Syst. (ICESC)*, Jul. 2020, pp. 143–148.

[12] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. S. Torr, "Riemannian walk for incremental learning: Understanding forgetting and intransigence," 2018, *arXiv:1801.10112*.

[13] L. Shan, W. Wang, K. Lv, and B. Luo, "Class-incremental learning for semantic segmentation in aerial imagery via distillation in all aspects," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615712.

[14] Y. Tai, Y. Tan, S. Xiong, and J. Tian, "Mine-distill-prototypes for complete few-shot class-incremental learning in image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5206013.

[15] J. Li et al., "Class-incremental learning network for small objects enhancing of semantic segmentation in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612920.

[16] C. Yu, B. Gong, M. Song, E. Zhao, and C.-I. Chang, "Multi-view calibrated prototype learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544713.

[17] Z. Lei, Y. Zeng, P. Liu, and X. Su, "Active deep learning for hyperspectral image classification with uncertainty learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[18] Q. Zhang, Q. Yuan, M. Song, H. Yu, and L. Zhang, "Cooperated spectral low-rankness prior and deep spatial prior for HSI unsupervised denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 6356–6368, 2022.

[19] Q. Zhang, Y. Zheng, Q. Yuan, M. Song, H. Yu, and Y. Xiao, "Hyperspectral image denoising: From model-driven, data-driven, to model-data-driven," *IEEE Trans. Neural Netw. Learn. Syst.*, no. 6, pp. 1–21, Jun. 2023, Art. no. 3278866.

[20] Q. Zhang, Y. Dong, Q. Yuan, M. Song, and H. Yu, "Combined deep priors with low-rank tensor factorization for hyperspectral image restoration," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[21] Y. Wang, Q. Zhu, H. Ma, and H. Yu, "A hybrid gray wolf optimizer for hyperspectral image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527713.

[22] Y. Wang et al., "Constrained-target band selection for multiple-target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6079–6103, Aug. 2019.

[23] C. Yu, C. Liu, H. Yu, M. Song, and C.-I. Chang, "Unsupervised domain adaptation with dense-based compaction for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 12287–12299, 2021.

[24] C. Yu, C. Liu, M. Song, and C. Chang, "Unsupervised domain adaptation with content-wise alignment for hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[25] M. Masana, X. Liu, B. Twardowski, M. Menta, A. D. Bagdanov, and J. van de Weijer, "Class-incremental learning: Survey and performance evaluation on image classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5513–5533, May 2023.

[26] X. Liu, M. Masana, L. Herranz, J. Weijer, A. Lopez, and A. D. Bagdanov, "Rotate your networks: Better weight consolidation and less catastrophic forgetting," in *Proc. Int. Conf. Pattern Recognit.*, Feb. 2018, pp. 2262–2268.

[27] J. Zhang et al., "Class-incremental learning via deep model consolidation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2019, pp. 1120–1129.

[28] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2018.

[29] Y. Wu et al., "Large scale incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 374–382.

[30] O. Ostapenko, M. Puscas, T. Klein, P. Jähnichen, and M. Nabi, "Learning to remember: A synaptic plasticity driven framework for continual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11313–11321.

[31] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "ICaRL: Incremental classifier and representation learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5533–5542.

[32] R. Aljundi, F. Babiloni, M. Elhoseiny, M. Rohrbach, and T. Tuytelaars, "Memory aware synapses: Learning what (not) to forget," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 144–161.

[33] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proc. Int. Conf. Mach. Learn.*, vol. 70, Mar. 2017, pp. 3987–3995.

[34] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Jul. 2016.

[35] M. Xu, Y. Zhao, Y. Liang, and X. Ma, "Hyperspectral image classification based on class-incremental learning with knowledge distillation," *Remote Sens.*, vol. 14, no. 11, p. 2556, May 2022.

[36] W. Zhao, R. Peng, Q. Wang, C. Cheng, W. J. Emery, and L. Zhang, "Life-long learning with continual spectral–spatial feature distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544214.

[37] S. D. Bhat, B. Banerjee, S. Chaudhuri, and A. Bhattacharya, "CILEA-NET: Curriculum-based incremental learning framework for remote sensing image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5879–5890, 2021.

[38] J. Bai et al., "Class incremental learning with few-shots based on linear programming for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 6, pp. 5474–5485, Jun. 2022.

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[39] Y. Zhao, L. Ha, H. Wang, and X. Ma, "Few-shot class incremental learning for hyperspectral image classification based on constantly updated classifier," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1376–1379.
[40] X. Lu, X. Sun, W. Diao, Y. Feng, P. Wang, and K. Fu, "LIL: Lightweight incremental learning approach through feature transfer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5611320.
[41] H. Tang, Y. Li, X. Han, Q. Huang, and W. Xie, "A spatial–spectral prototypical network for hyperspectral remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 167–171, Jan. 2020.
[42] B. Xi, J. Li, Y. Li, R. Song, and Q. Du, "Deep prototypical networks with hybrid residual attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3683–3700, 2020.
[43] F. Zhu, X.-Y. Zhang, C. Wang, F. Yin, and C.-L. Liu, "Prototype augmentation and self-supervision for incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5867–5876.
[44] G. Xu, Z. Liu, X. Li, and C. Loy, "Knowledge distillation meets self-supervision," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jun. 2020, pp. 1–11.
[45] G. E. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.
[46] M. Zhang, X. Zhao, W. Li, Y. Zhang, R. Tao, and Q. Du, "Cross-scene joint classification of multisource data with multilevel domain adaption network," *IEEE Trans. Neural Netw. Learn. Syst.*, no. 4, pp. 1–13, Apr. 2023, Art. no. 3262599.
[47] K. Gao, A. Yu, X. You, C. Qiu, B. Liu, and W. Guo, "Learning general-purpose representations for cross-domain hyperspectral images classification with small samples," *Remote Sens.*, vol. 15, no. 4, p. 1080, Feb. 2023.
[48] Y. Hu, Y. Huang, G. Wei, and K. Zhu, "Heterogeneous few-shot learning with knowledge distillation for hyperspectral image classification," in *Proc. 2nd Int. Conf. Consum. Electron. Comput. Eng. (ICCECE)*, Jan. 2022, pp. 601–604.
[49] L. Hu, W. He, L. Zhang, and H. Zhang, "Cross-domain meta-learning under dual-adjustment mode for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526416.
[50] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501813.
[51] Q. Chi, G. Lv, G. Zhao, and X. Dong, "A novel knowledge distillation method for self-supervised hyperspectral image classification," *Remote Sens.*, vol. 14, no. 18, p. 4523, Sep. 2022.
[52] F. M. Castro, M. J. Marín-Jiménez, N. Guil, C. Schmid, and K. Alahari, "End-to-end incremental learning," 2018, *arXiv:1807.09536*.
[53] F. Y. Wang, D. W. Zhou, H. J. Ye, and D. C. Zhan, "FOSTER: Feature boosting and compression for class-incremental learning," in *Proc. Eur. Conf. Comput. Vis.*, in Lecture Notes in Computer Science, 2022, pp. 398–414.



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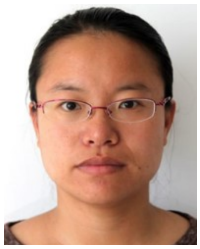
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